

QXU6018 Polymer Processing

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1. Introduction to Module

■ Overview

In spring semester, theory lecture.

In autumn semester, 6 lab experience.

■ Assessment

Activity	Score
Regular Homework Assignments	10%
Group Presentation	10%
Individual Coursework	40%
Exam in Week10	20%

2. Mechanical Behavior of Polymers

2.1. Viscoelastic Properties of Polymers

At a specific temperature and molecular weight, may behave as a liquid or a solid depend on time.

■ Comparison of Elastic vs. Viscous Behavior

Property	Elastic (Hookean)	Viscous (Newtonian)
Constitutive Equation	$\sigma = E\epsilon$	$\sigma = \mu\dot{\gamma}$
Response to Load	Instantaneous	Time-dependent
Energy	Stored	Dissipated
Recovery	Complete	None (permanent)
Rate Dependence	No	Yes

■ Stress Relaxation

The stress response to a unit constant strain:

$$E_e(t) = \sigma(t)/\epsilon(0)$$

2.2. Time-Temperature Superposition(TTS)

■ WLF Equation

$$\log a_T = \frac{-C_1(T - T_{ref})}{C_2 + T - T_{ref}}$$

C_1 and C_2 is material dependent constant.

If T_g is the ref temperature, $C_1 = 17.44$ and $C_2 = 51.6$.

■ Assumptions

1. The material should not undergo any chemical or physical changes as result of the temperature change.
2. No heterogeneity in sample, for example, blend.
3. Application in linear Viscoelastic regime only.

2.3. Dynamic Mechanical Test

■ General form

1

2.4. Fiber Reinforced Polymer

1

3. Rheology

3.1. What is Rheology

Rheology is the study of flow and deformation of all forms of matter.

$$N_1 = \tau_{xx} - \tau_{yy} = \psi_1(T, \dot{\gamma}) \dot{\gamma}_{xy}^2$$

$$N_2 = \tau_{yy} - \tau_{zz} = \psi_2(T, \dot{\gamma}) \dot{\gamma}_{xy}^2$$

■ Deborah Number

$$De = \frac{\lambda}{t_{process}}$$

...

3.2.